

ELDERLY CARE MONITORING SYSTEM WITH EMERGENCY ALERTS

Research Paper Review / Literature Survey

Submitted in the partial fulfilment for the Course of

CSA1012 SOFTWARE ENGINEERING

to the award of the degree of

BACHELOR OF ENGINEERING

IN

INFORMATION TECHNOLOGY

Submitted by

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Under the Supervision of

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November 2025



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DECLARATION

I, **MAHIMA (192521280)**, of the **INFORMATION TECHNOLOGY**, Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled '**ELDERLY CARE MONITORING SYSTEM WITH EMERGENCY ALERTS**' is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

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Date: 28-11-2025

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**ELDERLY CARE MONITORING SYSTEM WITH EMERGENCY ALERTS**” has been carried out by **Mahima (192521280)** under the supervision of **Dr. Rameswari** and is submitted in partial fulfilment of the requirements for the current semester of the B. Tech **INFORMATION TECHNOLOGY** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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ACKNOWLEDGEMENT

We would like to express our heartfelt gratitude to all those who supported and guided us throughout the successful completion of our Capstone Project. We are deeply thankful to our respected Founder and Chancellor, **Dr. N.M. Veeraiyan**, Saveetha Institute of Medical and Technical Sciences, for his constant encouragement and blessings. We also express our sincere thanks to our Pro-Chancellor, **Dr. Deepak Nallaswamy Veeraiyan**, and our Vice-Chancellor, **Dr. S. Suresh Kumar**, for their visionary leadership and moral support during this project.

We are truly grateful to our Director, **Dr. Ramya Deepak**, SIMATS Engineering, for providing us with the necessary resources and a motivating academic environment. Our special thanks to our Principal, **Dr. B. Ramesh** for granting us access to the institute's facilities and encouraging us throughout the process. We sincerely thank our Head of the Department, **Dr K. Malathi** for her continuous support, valuable guidance, and constant motivation.

We are especially indebted to our guides, **Dr. K. Anita Davamani** or their creative suggestions, consistent feedback, and unwavering support during each stage of the project. We also express our gratitude to the Project Coordinators, Review Panel Members (Internal and External), and the entire faculty team for their constructive feedback and valuable inputs that helped improve the quality of our work. Finally, we thank all faculty members, lab technicians, our parents, and friends for their continuous encouragement and support.

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Abstract

This literature survey reviews recent research on the assigned problem, “Elderly Care Monitoring System with Emergency Alerts,” using only research papers and scholarly review articles published between 2024 and 2026. The review focuses on wearable sensing, ambient and smart-home sensing, computer vision, radio-frequency sensing, Internet of Things (IoT) architectures, edge computing, deep learning, multimodal fusion, privacy-aware alerting and real-time emergency response.

Ten recent papers were selected from established scholarly publishers and journals, including Scientific Reports, Internet of Things, Measurement: Sensors, Sensors, Future Internet, Heliyon, Engineering Applications of Artificial Intelligence and Alexandria Engineering Journal. The reviewed evidence shows a shift from isolated fall detection toward multimodal, privacy-aware and edge-enabled monitoring. Recent systems report strong model performance, but the literature also exposes recurring weaknesses: small or simulated datasets, severe class imbalance, limited longitudinal validation, privacy concerns, sensor non-wear or failure, false alarms, and incomplete evaluation of the complete detection-to-caregiver-response chain.

The central research gap is therefore system-level rather than purely algorithmic. Existing studies demonstrate capable individual components, but there remains a need for a low-cost, elderly-centred, personalized and fault-tolerant platform that combines heterogeneous sensing, edge AI, calibrated risk assessment and reliable emergency escalation while protecting privacy. The proposed future direction is a multimodal monitoring architecture in which low-burden sensors continuously screen for risk, AI combines evidence across modalities, and an alert engine escalates only sufficiently credible events to caregivers.

Keywords: elderly care; emergency alerts; IoT; fall detection; smart home; wearable sensors; edge AI; multimodal fusion; privacy; real-time monitoring

1. Scope and Literature Selection

A strict publication-window rule was applied: every paper used as a selected study in this report was published in 2024, 2025 or 2026. Earlier papers are intentionally excluded from the selected-paper review and reference list. This ensures direct compliance with the requirement for recent literature.

The search strategy used combinations of the terms “elderly care monitoring,” “elderly fall detection,” “smart home,” “IoT healthcare,” “emergency alert,” “wearable sensor,” “ambient sensing,” “edge AI,” “multimodal fall detection,” “elderly health monitoring,” and “real-time warning system.” Priority was given to peer-reviewed research articles and systematic reviews with direct relevance to monitoring older adults, detecting emergencies or supporting caregiver response.

Year	Selected paper	Venue	Primary contribution
2024	Remote monitoring system for elderly falls using IoT and six-axis acceleration sensors	Measurement: Sensors	IoT + six-axis wearable fall monitoring
2024	Full-coverage unobtrusive health monitoring of elders at homes	Internet of Things	Multimodal ambient/physiological home monitoring
2024	In-home evaluation of Neo Care AI sound-based fall detection	Future Internet	Real-world home evaluation + user perceptions
2024	AI edge-computing wearable for elderly fall detection/prevention	Heliyon	Edge AI wearable + deep learning
2025	Privacy-aware fall detection and alert management in smart environments	Internet of Things	Thermal + wearable multimodal privacy-aware alerts
2025	LSTM anomaly detection in care-independent smart homes	CMES	SMOTE + LSTM anomaly detection
2025	Design of smart home integrating environmental sensing and AI safety monitoring	IJHISI	Environmental hazards + YOLOv5s
2026	Multimodal deep learning and wearable sensor fusion for fall risk	Scientific Reports	CNN-LSTM attention + graded real-time warnings

2026	Millimeter-wave multi-person fall detection	Scientific Reports	Contactless RF sensing validated with wearables/real life
2026	Lightweight IoT healthcare wearable for fall detection and ambient hazards	Procedia Computer Science	Wearable + environmental hazard alerts

2. Elderly Care Monitoring System Architecture

Figure 1 synthesizes the architecture implied by the recent literature: multiple sensing modalities feed local preprocessing and AI inference, followed by an emergency alert engine that communicates actionable information to caregivers. The architecture is deliberately multimodal because no single sensor type reliably covers motion, physiology, environmental hazards and context.

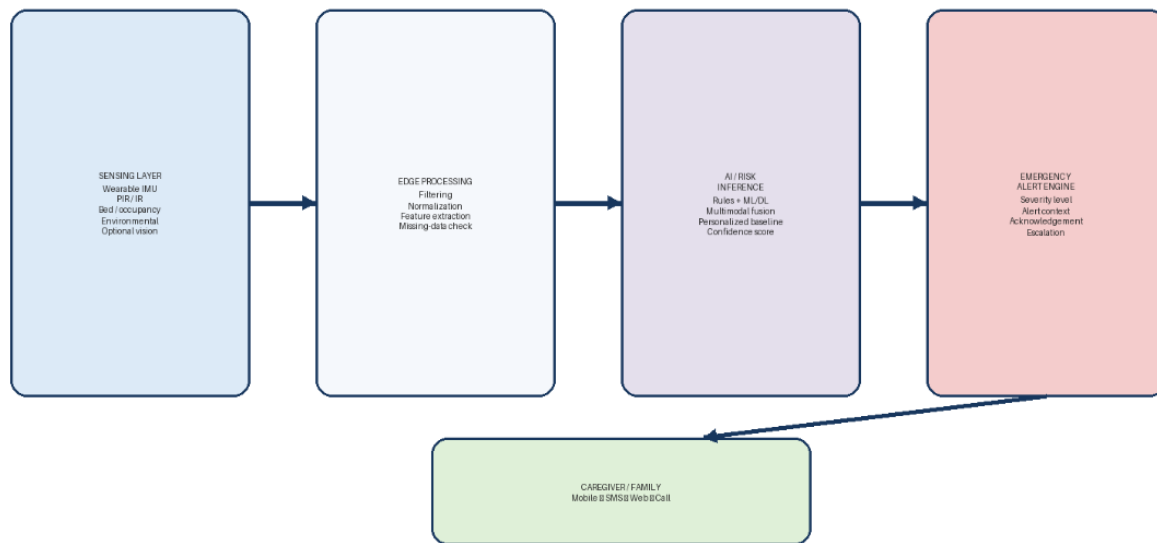


Figure 1. Elderly Care Monitoring System Architecture (original synthesis of the 2024–2026 literature).

3. Individual Research Paper Reviews

3.1 Research on the Remote Monitoring System for Falls in the Elderly Based on the Internet of Things and Six-Axis Acceleration Sensors (2024)

Objective: develop a remote elderly fall-monitoring system using IoT and six-axis acceleration sensing. Methodology: hardware and software modules were designed around real-time motion sensing and remote transmission. Technology: six-axis acceleration sensing, IoT communication and monitoring software. Dataset/evaluation: system-oriented experimental evaluation rather than a large clinical cohort. Major finding: the study demonstrates the feasibility of connecting wearable motion sensing to a remote monitoring architecture for rapid fall awareness. Limitation: the work is strongly centered on falls and system implementation, leaving broader health states, personalization and multimodal context less developed. Research gap: combine motion evidence with environmental and physiological signals and add intelligent alert confidence/escalation.

Source: (2024; Measurement: Sensors; DOI: 10.1016/j.measen.2024.101085)

3.2 Full-Coverage Unobtrusive Health Monitoring of Elders at Homes (2024)

Objective: achieve round-the-clock, unobtrusive monitoring for elderly people living alone. Methodology: a multimodal sensor array combines infrared, bed, sound and physiological sensors for activity, location, sleep and physiological assessment. Technology: smart sensors, multimodal sensing and intelligent data processing. Dataset: real-world data from ten older adults aged 65+ with 30-day activity records. Major finding: the system can recognize a broad range of home activities and support sleep and physiological monitoring without requiring continuous wearable use. Limitation: the cohort

is small and the study focuses on monitoring capability more than end-to-end emergency alert performance. Research gap: translate broad monitoring into calibrated emergency-risk scores and reliable caregiver escalation.

Source: (2024; Internet of Things, 26, 101182; DOI: 10.1016/j.iot.2024.101182)

3.3 In-Home Evaluation of the Neo Care Artificial Intelligence Sound-Based Fall Detection System (2024)

Objective: evaluate an AI sound-based fall detection system under real home conditions and examine user perceptions. Methodology: a 12-week in-home trial evaluated detection performance and user experience. Technology: AI-based acoustic sensing and home monitoring. Dataset/participants: 18 households involving adults aged 65+ at risk of falls; mean age 75.3 years and 67% female. Major finding: the study provides ecologically relevant evidence because the system was evaluated in participants' homes rather than only laboratory scenarios. Limitation: acoustic sensing can be affected by environmental noise and the study population is relatively small. Research gap: combine sound with non-audio ambient/wearable signals and evaluate alert burden and caregiver response over longer periods.

Source: (2024; Future Internet, 16(6), 197; DOI: 10.3390/fi16060197)

3.4 Development of Artificial Intelligence Edge Computing Based Wearable Device for Fall Detection and Prevention of Elderly People (2024)

Objective: build an AI edge-computing wearable for elderly fall detection and prevention. Methodology: deep-learning models including CNN, RNN, LSTM and GRU were investigated for activity recognition. Technology: wearable sensors, embedded/edge computing and deep learning. Dataset/evaluation: activity/fall recognition experiments were used to compare models and demonstrate feasibility. Major finding: edge processing can reduce dependence on cloud infrastructure and support near-real-time detection. Limitation: wearable-only monitoring can fail when the device is not worn, misplaced or uncharged, and the evaluation does not fully address multi-emergency detection. Research gap: use edge AI as one layer in a heterogeneous sensing architecture rather than as the sole source of evidence.

Source: (2024; Heliyon, 10(8), e28688; DOI: 10.1016/j.heliyon.2024.e28688)

3.5 Privacy-Aware Fall Detection and Alert Management in Smart Environments Using Multimodal Devices (2025)

Objective: combine privacy-aware sensing with fall detection and post-fall alert management. Methodology: wearable sensors and thermal cameras are combined, with edge-based computation for real-time detection; voice assistants and speakers support personalized post-fall interaction. Technology: thermal imaging, wearable sensing, CNN/RNN-style learning, edge computing and voice interaction. Evaluation: multimodal experiments compared sensor sources and machine-learning approaches. Major findings: thermal-image CNN detection achieved an F1-score above 0.98 and wearable-sensor models achieved about 0.93 F1; post-processing reduced false positives. Limitation: the system still focuses primarily on falls, and data availability is restricted. Research gap: generalize privacy-aware multimodal alerting to a wider set of elderly emergencies and validate long-term human response.

Source: (2025; Internet of Things, 30, 101526; DOI: 10.1016/j.iot.2025.101526)

3.6 LSTM Based Neural Network Model for Anomaly Event Detection in Care-Independent Smart Homes (2024)

Objective: detect anomalous events, especially falls, in care-independent smart homes while addressing imbalanced time-series data. Methodology: SMOTE was used to balance the dataset before training an

LSTM model to learn temporal dependencies. Technology: LSTM, SMOTE and smart-home time-series sensing. Major finding: the reported LSTM model achieved 84% accuracy and improved anomaly discrimination. Limitation: accuracy is insufficient for rare emergency events, and synthetic oversampling may not fully reproduce real emergency distributions. Research gap: use episode-level metrics, calibration and real-world longitudinal validation rather than accuracy alone.

Source: (2024; Computer Modeling in Engineering & Sciences, 140(3), 2689–2706; DOI: 10.32604/cmescs.2024.050825)

3.7 Design of a Smart Home System for Independent Elderly Integrating Environmental Sensing and AI-Based Safety Monitoring (2025)

Objective: support independent elderly living through environmental sensing and AI-based fall detection. Methodology: temperature/humidity, light, combustible-gas and air-quality sensors are combined with a YOLOv5s vision model and STM32F103 controller. Technology: IoT microcontroller, environmental sensors, camera-based object detection and remote monitoring. Evaluation: hardware/control logic was validated in Proteus and YOLOv5s was trained/evaluated on a PC. Major finding: the architecture can address both environmental hazards and falls. Limitation: hardware validation is not equivalent to long-term real-home clinical validation; camera privacy is also a concern. Research gap: move from prototype validation to real homes and add privacy-preserving edge inference.

Source: (2025; International Journal of Healthcare Information Systems and Informatics, 20(1); DOI: 10.4018/IJHISI.389199)

3.8 Intelligent Fall Risk Prediction and Real-Time Warning System for Elderly Care Based on Multimodal Deep Learning and Wearable Sensor Fusion (2026)

Objective: continuously estimate fall risk using multiple wearable modalities rather than periodic assessment or one sensor. Methodology: triaxial acceleration, gyroscope, insole pressure and wrist PPG were fused using a CNN-LSTM backbone with eight-head attention; a rule engine mapped continuous risk into four alert tiers. Dataset: 120 community-dwelling older adults over six months; held-out test set of 18 participants with 12,540 windows and six fall events. Major findings: 94.2% overall accuracy, AUC 0.967 and average smartphone latency of 138 ms. The paper explicitly reports extreme class imbalance and only 6.25% window-level PPV. Limitation: the small number of actual falls makes sensitivity estimates unstable and the later deployment was not a clinical efficacy trial. Research gap: episode-level calibrated alerts and larger prospective validation are needed.

Source: (2026; Scientific Reports; DOI: 10.1038/s41598-026-56750-9)

3.9 Millimeter-Wave Technology for Multi-Person Fall Detection Validated Through Wearable Sensors and Real-Life Scenarios (2026)

Objective: develop contactless multi-person fall detection using millimeter-wave sensing and validate it against wearable and real-life scenarios. Methodology: RF/mmWave sensing is used to detect human movement without cameras or worn devices. Technology: millimeter-wave radar sensing, signal processing and AI-based classification. Evaluation: validation includes wearable references and real-life scenarios. Major finding: the study demonstrates the feasibility of privacy-preserving contactless sensing in multi-person environments. Limitation: RF sensing can be affected by room geometry, occlusion, multipath effects and deployment configuration. Research gap: combine RF sensing with ambient and wearable evidence and develop robust cross-home calibration.

Source: (2026; Scientific Reports, 16, 8859)

3.10 A Lightweight IoT Healthcare Wearable for Fall Detection and Ambient Hazard Sensing (2026)

Objective: provide continuous elderly monitoring through a lightweight IoT wearable capable of identifying falls and environmental safety risks. Methodology: multiple sensors collect vital/safety information, with detected events transmitted to a web server and presented through a remote dashboard. Technology: IoT wearable, fall sensing, environmental/thermal sensing, web dashboard and remote alerts. Evaluation: proof-of-concept implementation demonstrates real-time monitoring and alert generation. Major finding: combining wearable health data with ambient hazard information can improve caregiver awareness and responsiveness. Limitation: proof-of-concept status means broader clinical and longitudinal validation remains necessary. Research gap: develop fault-tolerant, personalized and privacy-preserving edge processing with validated caregiver-response metrics.

Source: (2026; Procedia Computer Science, 280, 406–413; DOI: 10.1016/j.procs.2026.04.052)

4. Literature Evolution / Technology Taxonomy

The 2024–2026 literature can be organized into four interacting technology families: wearable sensing, ambient/IoT sensing, computer vision/RF sensing, and multimodal edge intelligence. Recent work increasingly combines these families instead of treating them as mutually exclusive.

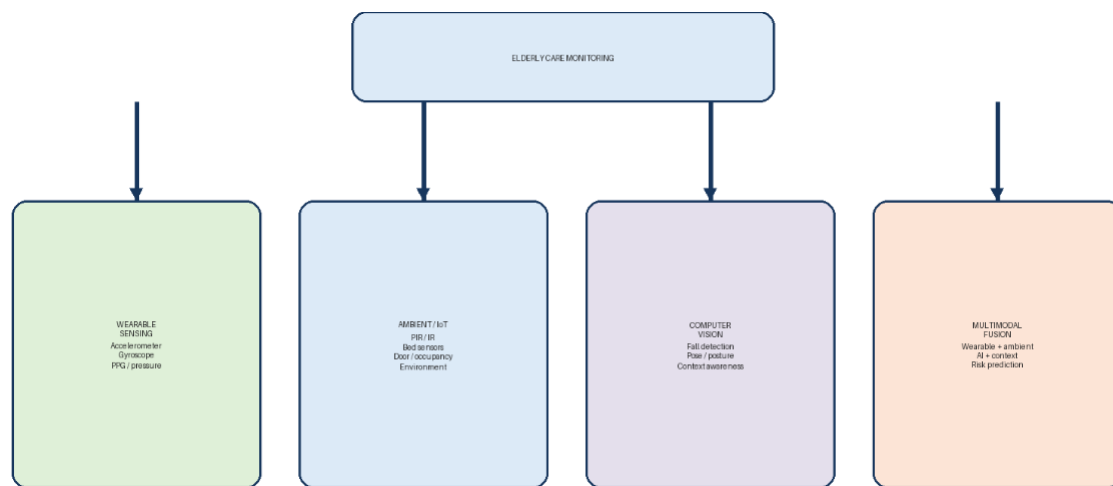


Figure 2. Technology taxonomy synthesized exclusively from the 2024–2026 selected literature.

5. Comparative Analysis

The selected studies cannot be ranked using accuracy alone because they solve different tasks, use different datasets and measure performance at different levels. A strong literature comparison therefore includes sensing burden, privacy, edge capability, real-world validation, alert support and major limitations alongside reported performance.

Study (year)	Sensing approach	AI / processing	Dataset evaluation	Key reported result	Alert relevance	Main limitation
Remote IoT fall monitoring (2024)	Six-axis wearable	IoT + motion analysis	System evaluation	Remote fall-monitoring feasibility	High	Fall-centric; limited broader context
Full-coverage home monitoring (2024)	IR + bed + sound + physiology	Multimodal processing	10 older adults; 30 days	Broad activity/health coverage	Medium–High	Small cohort
Neo Care (2024)	Sound / acoustic	AI detection	18 households; 12 weeks	In-home validation	High	Noise + small sample
Edge wearable (2024)	Wearable motion	CNN/RNN/LSTM/GRU	Activity/fall experiments	Edge AI feasibility	High	Non-wear and limited emergency types
Privacy-aware multimodal (2025)	Thermal wearable +	CNN/ML + edge	Multimodal experiments	Thermal F1 >0.98; wearable F1 ~0.93	Very high	Limited long-term validation
LSTM smart home (2024)	Smart-home time series	SMOTE + LSTM	Anomaly dataset	84% accuracy	High	Accuracy alone; synthetic balancing
AI smart home (2025)	Environment + camera	YOLOv5s + MCU	Proteus + PC evaluation	Integrated hazard + fall design	Very high	Limited real-home hardware testing

Multimodal fall risk (2026)	4 wearable modalities	CNN-LSTM + attention	120 older adults; 6-month study	94.2% accuracy; AUC .967; 138 ms	Very high	6 actual falls; 6.25% window PPV
mmWave fall detection (2026)	RF/mmWave	AI/signal processing	Wearable + real-life validation	Contactless multi-person feasibility	High	Room/deployment sensitivity
Lightweight IoT wearable (2026)	Wearable + ambient hazards	IoT + dashboard	Proof-of-concept	Real-time alerts and remote monitoring	Very high	Needs larger validation

6. Emergency Alert Workflow

Recent research increasingly treats alerting as a system function rather than merely the output of a classifier. The proposed workflow below separates anomaly detection from risk confirmation and caregiver escalation. This is important because rare-event systems can generate false positives if every anomalous sensor window is immediately treated as an emergency.

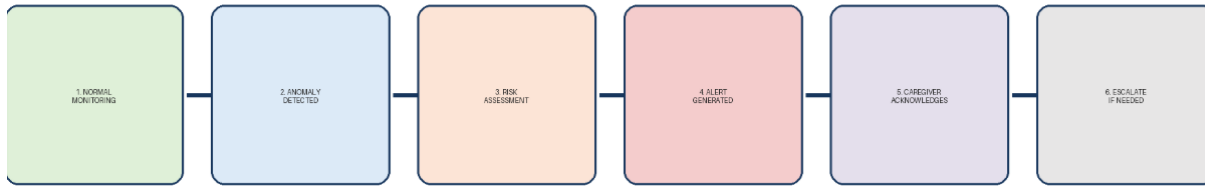


Figure 3. Recommended emergency alert workflow based on limitations identified across 2024–2026 studies.

7. Critical Analysis: Limitations and Challenges

7.1 Dataset and validation limitations

Several studies use small cohorts, controlled experiments, simulations or proof-of-concept deployments. The 2026 multimodal study is stronger because it includes 120 older adults over six months, but only six fall events occurred in the held-out test set. This illustrates why real emergency events are difficult to collect and why model evaluation must be designed around rare-event statistics.

7.2 Class imbalance and misleading accuracy

Emergency events are rare compared with normal daily activity. Consequently, high accuracy can coexist with weak positive predictive value. The 2026 multimodal study explicitly reports 94.2% accuracy and AUC 0.967 while also reporting only 6.25% window-level PPV. This is a critical lesson: false alarms and missed emergencies must be measured at the event/episode level.

7.3 Wearability and user burden

Wearable systems provide rich motion and physiological signals but depend on adherence, charging and correct placement. An elderly-care system should therefore treat missing wearable data as a normal operating condition and maintain safety using ambient sensing where possible.

7.4 Privacy

Cameras provide contextual information but can be unacceptable in private spaces. Thermal and RF sensing provide promising alternatives. Edge processing can further reduce the need to transmit raw sensor streams or video.

7.5 Environmental robustness

Ambient, acoustic, RF and vision systems are affected by room layout, noise, lighting, multipath, occlusion and sensor placement. Cross-home calibration and fault detection are necessary for reliable deployment.

7.6 Alert fatigue and caregiver response

An alert has value only when it produces an appropriate response. Recent work on alert management highlights the importance of personalized responses, post-event assessment and human involvement.

Future systems should therefore measure acknowledgement and escalation latency, not only classification accuracy.

8. Research Gap Identification

The strongest research gap emerging from the 2024–2026 literature is the fragmentation between sensing, prediction and emergency response. Individual papers demonstrate strong components—edge AI, multimodal sensing, smart-home monitoring, privacy-aware thermal detection or real-time warning—but few evaluate a complete, low-cost and personalized platform under realistic failure conditions.



Figure 4. Research gap identified by synthesizing the 2024–2026 evidence.

Gap 1 — End-to-end integration: Most studies optimize one sensing modality or one detection task. A complete elderly-care system needs sensing, inference, alert delivery, acknowledgement and escalation in one evaluated pipeline.

Gap 2 — Personalized risk assessment: Static thresholds do not adequately represent individual routines, mobility and baseline physiology. Personal baselines and adaptive thresholds are needed.

Gap 3 — Rare-event evaluation: Emergency events are sparse. Event-level precision, recall, false alarms/day, missed emergencies, calibration and PR-AUC should accompany accuracy and ROC-AUC.

Gap 4 — Robustness to missing data: Real users may remove wearables, sensors may fail, batteries may die and networks may disconnect. Systems need graceful degradation and fail-safe alert pathways.

Gap 5 — Privacy-preserving multimodality: There is a need to combine rich contextual sensing with privacy constraints using edge inference, thermal/RF sensing and selective activation of cameras.

Gap 6 — Human-centred alert escalation: The system should identify who should be contacted, request acknowledgement, avoid duplicate alerts and escalate only when necessary.

Gap 7 — Longitudinal validation: Short pilots and simulations cannot fully capture technology fatigue, changing routines, seasonal conditions and real caregiver workflows.

Gap 8 — Multi-emergency coverage: Falls are dominant in the literature, but elderly emergencies can also involve abnormal vital signs, gas/air hazards, prolonged inactivity and wandering. A broader risk engine is needed.

9. Proposed Research Direction for the Assigned Problem

Based on the recent literature, the proposed direction is a privacy-aware, low-cost, multimodal elderly monitoring platform with an edge-based Emergency Alert and Escalation Engine. The platform should

not replace caregivers; it should reduce detection and response time while filtering low-confidence anomalies.

Layer	Proposed components	Purpose
Sensing	Wearable IMU/PPG; PIR/IR; bed/occupancy; temperature; gas/air quality; optional thermal/RF/vision	Collect complementary evidence
Edge processing	Filtering; normalization; missing-data detection; event windows	Low latency and reduced raw-data transmission
AI inference	LSTM/CNN/Transformer or lightweight multimodal model + rules	Detect and classify risk
Personalization	Individual activity baseline; adaptive thresholds; confidence calibration	Reduce false alarms
Alert engine	Severity tiers; context; acknowledgement timer; escalation	Convert predictions into action
Communication	Wi-Fi/Bluetooth/cellular/backup local alarm	Resilient alert delivery
Caregiver interface	Mobile/web dashboard; event history; status	Support rapid human response
Security	Encryption; authentication; role-based access; audit log	Protect sensitive information

9.1 Recommended Evaluation Metrics

Dimension	Recommended measures
Detection	Sensitivity/recall, specificity, precision/PPV, F1, ROC-AUC, PR-AUC, confusion matrix
Rare-event safety	False alarms/day, missed emergencies, event-level PPV, episode sensitivity
Latency	Sensor-to-detection, detection-to-alert, alert-to-acknowledgement, escalation latency
Reliability	Uptime, sensor-dropout recovery, battery life, network outage tolerance
Usability	SUS, adherence/wear time, caregiver workload, satisfaction
Privacy	Raw-data retention, on-device processing ratio, access-control/audit events
Scalability	Users/devices supported, message throughput, compute and storage cost

10. Future Research Directions

Develop multimodal edge AI that fuses wearable, ambient, environmental and privacy-preserving RF/thermal evidence.

Use personalized behavioural and physiological baselines so that the system detects meaningful deviations rather than relying on universal thresholds.

Adopt rare-event learning and calibrated episode-level decision rules to reduce false alarms.

Design graceful degradation so monitoring continues when one sensor, wearable or network link becomes unavailable.

Use privacy-preserving sensing and local inference, retaining only the minimum information required for an alert.

Implement human-in-the-loop alert escalation with acknowledgement, cancellation, secondary-caregiver escalation and audit history.

Build longitudinal datasets with naturally occurring events and subject-independent external validation.

Expand beyond falls to abnormal vital signs, environmental hazards, prolonged inactivity and wandering/anomalous behaviour.

Evaluate operational outcomes such as caregiver response time and alert burden in addition to AI accuracy.

Design modular APIs so additional IoT sensors and healthcare devices can be integrated without redesigning the complete platform.

11. Conclusion

The 2024–2026 literature demonstrates rapid progress toward intelligent elderly-care monitoring. Recent work covers IoT fall monitoring, unobtrusive multimodal home sensing, AI acoustic detection, edge-computing wearables, privacy-aware thermal/wearable fusion, smart-home anomaly detection, environmental hazard sensing, multimodal deep learning, millimeter-wave monitoring and lightweight IoT wearables.

The research gap is therefore an end-to-end, personalized and resilient elderly-care platform that combines heterogeneous sensors with edge intelligence and human-centred emergency escalation. The strongest future system should be evaluated not only by whether it detects an event, but by whether it reliably communicates a meaningful alert, receives acknowledgement and supports timely intervention. This direction directly addresses the limitations identified across the selected 2024–2026 literature.

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